

Binary Tree Traversals (Inorder, Preorder, Postorder, Level Order) Data Structures

> **Definition:** Tree Traversal is the systematic process of visiting (reading, processing, or printing) every node in a tree data structure exactly once.

1. Detailed Technical Explanation

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1. Depth-First Traversals (DFS):

1. **Preorder Traversal (Root -> Left -> Right):**

- Vis

2. Breadth-First Traversal (BFS / Level Order):

- Visits nodes level-by-level from top to bottom, and left to right at each level using a **FIFO Queue**.

- *Ex

2. Complete Executable C Program for Traversals

```
```c
```

#includ

```
struct Node {
```

int data;

```
struct Node* createNode(int val) {
```

struct l

```
// 1. Inorder Traversal: Left -> Root -> Right
```

```
void in
```

```
// 2. Preorder Traversal: Root -> Left -> Right
```

```
void pr
```

```
// 3. Postorder Traversal: Left -> Right -> Root
```

```
void po
```

```
int main() {
```

```
// Cons
```

```
printf("Preorder : "); preorder(root); printf("\n");
```

```
printf("
```

```

```

### 3. Reconstructing Unique Binary Tree from Traversals

- A unique binary tree can be constructed if and only if **\*\*INORDER\*\*** is given along with either **\*\*PREORDER\*\*** or **\*\*POSTORDER\*\***.

- Ino

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#### 4. Must-Write Points for Exams

- Inorder traversal of BST gives non-decreasing sorted order.

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#### 5. Quick Recall Flow

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Preord